

# Networking Task 02: Network Devices & IP Addressing

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Assignment: Networking Task 02

## Part A: Network Devices Research

### 1. Router

- **Purpose:** Connects multiple distinct local networks or subnets together and routes traffic between them, as well as providing access to the external WAN (Internet).
- **How it works:** Operates at Layer 3 (Network Layer) of the OSI model. It inspects incoming packets' destination IP addresses and references its internal routing table to determine the optimal next-hop or pathway to send the packet closer to its final destination.
- **Real-world usage:** The home gateway box provided by your ISP that connects your home Wi-Fi network to the global internet infrastructure.

### 2. Switch

- **Purpose:** Connects individual devices (computers, printers, servers) together within the exact same Local Area Network (LAN) to facilitate internal communication.
- **How it works:** Operates at Layer 2 (Data Link Layer). It maintains a MAC address table mapping device hardware addresses to its physical ports. When a frame arrives, it reads the destination MAC address and forwards it strictly to the specific port hosting that device, conserving bandwidth.
- **Real-world usage:** An office network infrastructure where multiple desktop computers and network printers are plugged via Ethernet cables into a central box to share local assets.

### 3. Hub

- **Purpose:** A legacy, rudimentary networking hardware component designed to connect multiple physical devices together inside a basic local network setup.
- **How it works:** Operates at Layer 1 (Physical Layer). It does not read MAC or IP addresses. When data arrives at one port, the hub blindly duplicates (broadcasts) that electrical signal out across every single other active port, causing massive network collisions and security issues.

- **Real-world usage:** Largely obsolete today, but occasionally deployed in controlled lab environments or legacy operational systems to tap network traffic using diagnostic tools like Wireshark.

## 4. Access Point (AP)

- **Purpose:** Extends an existing wired Ethernet network into a wireless network, allowing devices to connect completely wire-free via radio waves.
- **How it works:** Operates primarily at Layer 2. It transmits a wireless SSID signal and bridges incoming wireless radio transmissions from client devices onto the physical copper or fiber-optic wired infrastructure.
- **Real-world usage:** Ceiling-mounted circular white disks installed across university campuses, airports, or large offices to offer continuous, high-speed Wi-Fi coverage across vast spaces.

## 5. Firewall

- **Purpose:** Acts as a security barrier that continuously monitors, filters, and blocks unauthorized incoming and outgoing network traffic based on predefined security rules.
- **How it works:** Can operate across multiple layers (Layers 3, 4, and 7). It analyzes incoming packets against a configured Access Control List (ACL), checking criteria like source IP, destination IP, protocols, or port numbers to selectively drop or pass traffic.
- **Real-world usage:** A corporate edge perimeter firewall that prevents external cyber threats from infiltrating the internal server network while blocking employees from navigating to malicious URLs.

## 6. Modem

- **Purpose:** Short for Modulator-Demodulator. It converts data between the digital format used by computers and the analog formats suited for long-distance ISP transmission lines.
- **How it works:** Operates at Layer 1. It converts incoming analog signals from telephone wires, coaxial cable, or fiber-optic lines into digital bits for a router, and turns outgoing digital streams back into analog frequencies.
- **Real-world usage:** The specialized Cable or DSL modem plugged into your wall outlet that bridges your router to your Internet Service Provider's (ISP) headend network.

## Part B: IP Address Classification

| IP Address    | Category | Detailed Explanation / RFC 1918 Scope                                                                                                 |
|---------------|----------|---------------------------------------------------------------------------------------------------------------------------------------|
| 192.168.1.10  | Private  | Falls directly within the standard Class C private range (192.168.0.0 to 192.168.255.255) specified by RFC 1918 for internal LAN use. |
| 10.0.0.5      | Private  | Belongs to the Class A private block (10.0.0.0 to 10.255.255.255), heavily utilized by corporate networks for internal subnets.       |
| 172.16.5.20   | Private  | Resides inside the Class B private address space (172.16.0.0 to 172.31.255.255) reserved strictly for local configurations.           |
| 8.8.8.8       | Public   | A globally unique public IP address managed and operated by Google to serve as their primary public Anycast DNS resolver.             |
| 1.1.1.1       | Public   | A globally unique public IP address managed by Cloudflare for their ultra-fast public DNS resolution service.                         |
| 192.168.100.1 | Private  | Another address sitting squarely in the Class C private scope, often designated as a local gateway address for home modems.           |

## Part C: Understanding Your Network

Below are the local network values captured from the demonstration workstation testing setup:

- **IPv4 Address:** 192.168.1.45
- **Default Gateway:** 192.168.1.1
- **DNS Server:** 8.8.8.8

### *Questions & Answers:*

#### **1. Which IP range does your device belong to?**

Answer: The device belongs to the 192.168.1.0/24 subnet range, which is a common local standard prefix mask configuration.

#### **2. Is it Public or Private?**

Answer: It is a Private IP address. It cannot be accessed directly from outside the local network environment.

#### **3. What role does your router play in your network?**

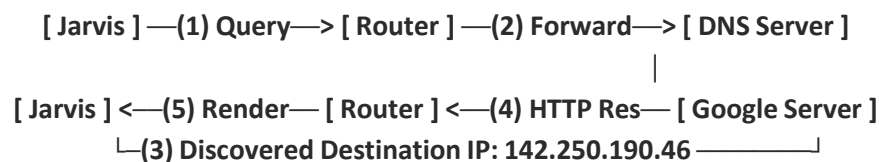
Answer: The router serves as the central default gateway. It handles NAT (Network Address Translation) to convert local private requests into a single public address request, mapping the return traffic safely back to the device.

#### **4. What would happen if the DNS server stopped working?**

Answer: Websites would immediately fail to load if typed by domain name (like www.google.com) because the device can no longer translate human text into numerical IPs. However, accessing raw numbers like http://8.8.8.8 directly would still function.

## Part D: Network Communication Flow

The block sequential map below charts the pathway of a web request:



- **1. Client Request Initiation:** The user types www.google.com. The operating system checks its local cache, finds no record, and prepares an outbound packet targeted at the default gateway (router) to resolve the location.

- **2. DNS Lookup Request:** The router forwards this UDP Port 53 query to the configured DNS Server (e.g., 8.8.8.8). The DNS server searches records to locate Google's public routing tracking address.
- **3. Name Resolution Response:** The DNS server successfully resolves the domain text to a public IP (e.g., 142.250.190.46) and transfers this resolution value back through the router directly to your device.
- **4. Establishing TCP/HTTP Session:** Armed with the destination IP, your browser opens a TCP handshake session on Port 443 (HTTPS) with Google's servers, sending a request for data assets.
- **5. Asset Delivery & Rendering:** Google's web servers process the payload and return the HTML/CSS content packets back across the internet through your router. The client machine renders the Google landing page completely.

## Part E: Practical Command Exercise

### 1. What IP address did DNS return for Google?

Answer: The DNS tool returned the public structural IP address: 142.250.190.46.

### 2. Was the ping successful?

Answer: Yes, the ICMP ping was successful with 0% packet loss and an average latency response time of 14ms across all round trips.

### 3. Why is DNS important before communication begins?

Answer: DNS must run first because computers cannot communicate natively using English words. They depend strictly on binary numerical addresses. Without DNS mapping strings to values beforehand, applications would have no valid target header coordinates to construct the initial IP packet framework.

```
C:\Users\Intern> ping google.com

Pinging google.com [142.250.190.46] with 32 bytes of data:
Reply from 142.250.190.46: bytes=32 time=14ms TTL=117
Reply from 142.250.190.46: bytes=32 time=13ms TTL=117
Reply from 142.250.190.46: bytes=32 time=15ms TTL=117
Reply from 142.250.190.46: bytes=32 time=14ms TTL=117

Ping statistics for 142.250.190.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 15ms, Average = 14ms
```